

```
In [4]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [5]: data = sns.load_dataset('iris')
```

```
In [6]: data.head()
```

```
Out[6]:
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

```
In [7]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype  
---  -
0   sepal_length    150 non-null   float64
1   sepal_width     150 non-null   float64
2   petal_length    150 non-null   float64
3   petal_width     150 non-null   float64
4   species         150 non-null   object  
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

```
In [8]: data.describe()
```

```
Out[8]:
```

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333
std	0.828066	0.435866	1.765298	0.762238
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

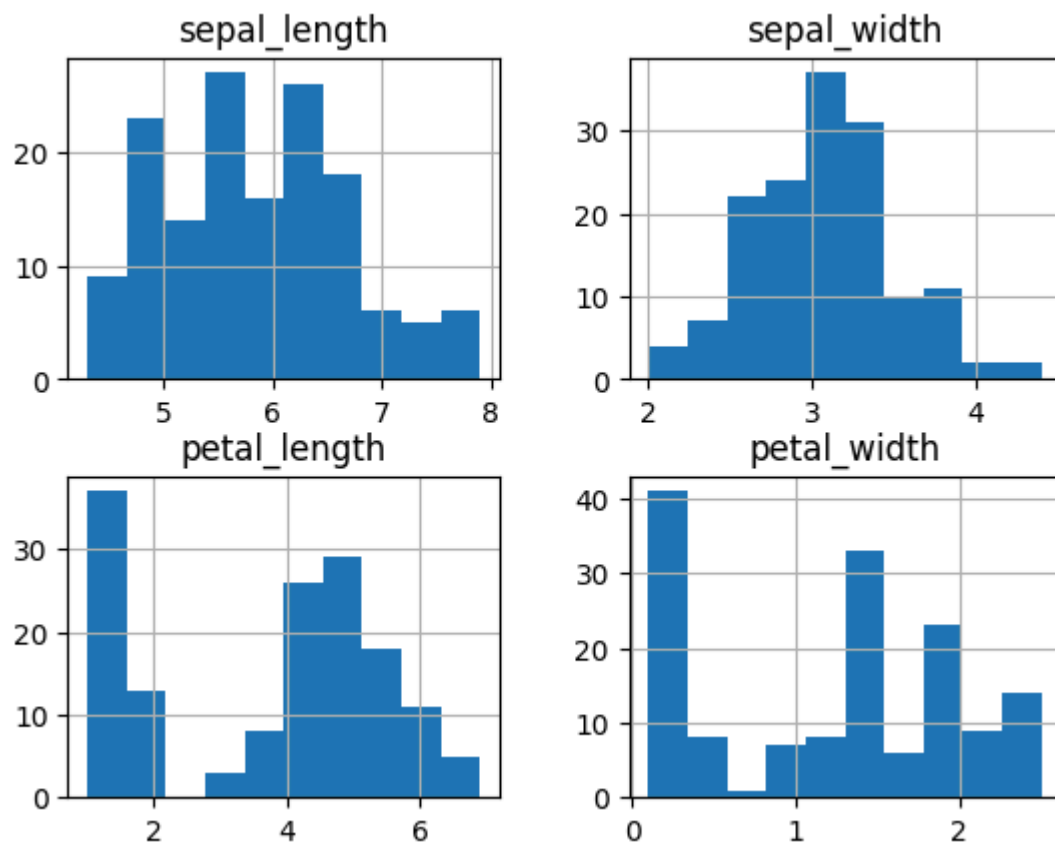
```
In [9]: data.isnull().sum()
```

```
Out[9]: sepal_length    0
sepal_width    0
petal_length    0
petal_width    0
species        0
dtype: int64
```

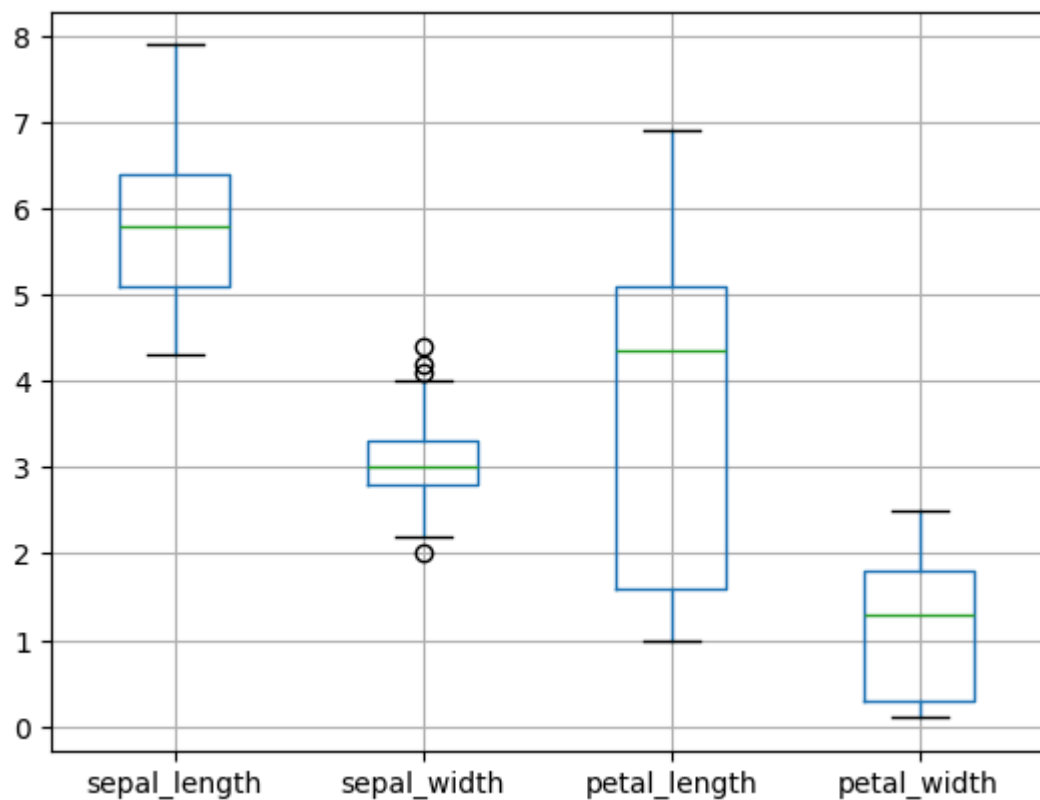
```
In [10]: data.shape
```

```
Out[10]: (150, 5)
```

```
In [11]: data.hist()  
plt.show()
```



```
In [13]: data.boxplot()  
plt.show()
```



```
In [14]: Q1 = data.drop(columns=['species']).quantile(0.25)
        Q3 = data.drop(columns=['species']).quantile(0.75)
        IQR = Q3 - Q1
```

```
In [18]: outliers = ((data.drop(columns=['species']) < (Q1 - 1.5 * IQR)) | (data.drop(columns=['species']
        outlier_counts = outliers.sum())
```

```
In [19]: print(outlier_counts)
```

```
sepal_length    0
sepal_width     4
petal_length    0
petal_width     0
dtype: int64
```

```
In [20]: df_cleaned = data[~outliers.any(axis=1)]
```

```
In [21]: print("Dataset before removing outliers:", data.shape)
        print("Dataset after removing outliers:", df_cleaned.shape)
```

```
Dataset before removing outliers: (150, 5)
Dataset after removing outliers: (146, 5)
```